

Promoting Financial Inclusion: Do Unconditional E-Money Transfers Work?

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Author Note

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Abstract

This research paper evaluates the impact of social assistance payments in the form of mobile money on financial inclusion. Through digital wallets, we exploit the disbursement of monetary support for vulnerable people in Colombia. Using a Regression Discontinuity Design (RDD) and administrative data on transactions, we test the effects of mobile money aid on transaction use, such as wire money transfers and cash-outs, and end-of-the-month balances. We find that (i) the program increased by 1.4 p.p. the average probability of using wire money transfers in the period analyzed. This represents an increase of 24% compared to the control group's average use of wire money transfers. (ii) The probability of cashing money out decreases by 76.6% when comparing the first semester to the fifth semester of continuous use. Finally, (iii) we find that eligible users increased their end-of-the-month balance by 27.6% compared to the non-eligible group's average end-of-the-month deposit account balance.

JEL Classification: G50; O33; O35

Keywords: financial inclusion, mobile money, digital wallets, regression discontinuity

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1. Introduction

The COVID-19 pandemic has had devastating effects on people's lives all over the world. Individuals' income and financial stability absorbed a big part of the economic shock caused by the sanitary emergency. The World Bank estimated that, after one year of the onset of the pandemic, more than 97 million people fell under the poverty line, living on less than 1.90 USD per day (Mahler et al., 2021). No modern government had faced such an enormous social crisis before, which forced them to take drastic and innovative measures to help their citizens.

Governments worldwide implemented Government-to-Person (G2P) payment programs in a matter of weeks to provide financial aid for their citizens. This aid was delivered as electronic money (e-money) in people's cell phones to comply with the social distancing measures. Furthermore, the use of e-money turned out to be a powerful mechanism to increase financial inclusion in communities with no prior access to the traditional financial system.

This paper aims to disentangle the effects of e-money on financial inclusion. For this purpose, we study the case of Ingreso Solidario (IS), a financial aid program implemented in Colombia right after the pandemic's start. It delivered e-money aid to vulnerable individuals. We hypothesize that IS increased financial inclusion by facilitating basic financial transactions and saving accounts using e-money in digital wallets.

To test our hypothesis, we gathered more than 3.5 million observations of monthly administrative data on individuals' transactions and saving balances. Additionally, we exploit the socioeconomic score SISBEN as the assignment variable to estimate the Intention-to-treat (ITT) and the Local Average Treatment Effect (LATE) under a regression discontinuity design (RDD).

Furthermore, our study uses high-quality data to contribute new evidence on e-money's effects on financial inclusion. We are interested in evaluating one of the initial slabs of what is known as the Theory of Change flow diagram for financial inclusion depicted by Duvendack & Mader (2020). In particular, this study focuses on a new technology initiative for financial inclusion- e-money- and its effects on enhancing the usage of three primary financial services: wire money transfers, cash-out, and deposits.

Our study finds relevant evidence of the effects of unconditional money transfers on financial inclusion for the five-semester period analyzed. We find that IS's ITT effect is, on average, a 1.4 p.p. increase in the probability of using wire money transfers. This effect represents a 24% growth in the probability of usage of this service compared to the average usage of the control group. In the case of cash-out transactions, we find that IS-treated individuals tend to cash their money out more than the control group. However, as expected for the technology adoption process, we find that the probability of cashing out decreases by 76.6% when compared to the probabilities of the first and the fifth semester of usage. Additionally, the end-of-the-month balance amount increases on average by 27.6% compared to the average control group balance.

The paper is structured into seven sections. The first section is the current introduction. The second section provides the background and context of G2P payments, e-money usage, and IS implementation in Colombia. The third section briefly discusses the literature reviewed on e-money's effects on financial inclusion. The following section describes in detail the data used for this study. The fifth section explains the RD design methodology employed to test the paper's hypothesis. Then, the results section discusses the main findings of this study. Finally, the last section concludes with the central ideas of this paper.

2. Background and Context

2.1 COVID-19 pandemic effects on financial stability

Colombia confirmed its first case of COVID-19 on March 6th. The World Health Organization declared the COVID-19 outbreak a global pandemic on March 11th, 2020. Then, the Colombian Government declared national quarantine starting on March 23rd. This decision forced millions of households to stay home, and many lost all their income because they were not able to work from home.

The pandemic economic crisis is projected to be the largest one seen in more than a century (World Bank, 2022). As a result, 90% of the global countries reported economic output contraction. In terms of poverty, it was estimated that 635 million individuals lived in extreme poverty before the pandemic. After the pandemic's start, this figure increased to 732 million individuals, with a total rise of approximately 97 million people induced into extreme poverty. In particular, Colombia, a country of about 50 million inhabitants, registered in 2020 more than 21 million individuals living in monetary poverty, 3.6 million more than in 2019 (DANE, 2022).

Governments around the globe have implemented several types of policies aiming to contain the spread of the virus, leveraging lockdown measures that restricted mobility and affected specific economic sectors. These policies mainly affected low-income workers with less education, minorities, immigrants, and women.

2.2 G2P payments implementation

G2P programs were implemented worldwide in a relatively short time to help people cope with the new challenges presented at the start of the pandemic. Leveraging Fintech's comparative advantage in reducing exposure to the virus and capillarity in isolated areas, governments worldwide relied on these institutions on top of banks with robust digital platforms to deploy

economic relief programs in populations with significant welfare deterioration due to the pandemic. These programs aimed to reduce the effects of the pandemic on households with non-contributory cash transfers to support consumption for the most vulnerable population.

Since it was essential to comply with the social distancing measures to avoid the COVID-19 virus contagion, much of the economic aid was delivered electronically. In the case of cash transfers aid, they were transferred to beneficiaries' savings accounts. Governments took advantage of the high penetration of mobile phone subscriptions, 70% according to GSMA (2020), to deliver financial help through digital wallets that can be accessed from a person's cell phone and use the resources as e-money.

Thus, one year after the pandemic was declared, the number of government assistance programs went from 103 in March 2020 to 1,841 in May 2021, reaching more than 1.5 billion people. Many of these programs used digital infrastructures to deliver funds quickly without physical contact, implying a significant growth in the number of digital savings accounts and wallets (Center for Financial Inclusion, 2022).

Calculations by Gentilini et al. (2020) at the World Bank show that approximately 50% of the population in North America, East Asia, and the Pacific received at least one unconditional transfer, while this figure drops to 19% in Eastern Europe and Central Asia and 10% in Africa. Of the 202 cash transfer programs studied, 58% were disbursed through digital payments and 22% through manual and digital payments. Interestingly, low-income countries display the highest share of digital payment deployment with 63%. In many countries, government support was vital to stabilizing consumption after the shock: The size of cash transfer programs represents, on average, 30% of monthly GDP per capita for all regions and 25% for Europe and Central Asia. In monetary

value, monthly transfers averaged USD 256, ranging from USD 525 in high-income countries to USD 42 in low-income countries.

In Latin America, 199 monetary support programs were implemented, and 56% were cash transfer programs (Cejudo et al., 2021). Other types of financial support programs were salary subsidies (13%), food stamps (12%), and tax exemptions (5%).

2.3 E-money: What is it?

Electronic money (e-money) is the electronic alternative to cash. It is stored in digital accounts provided by digital software (Apps) known as digital wallets. Users can perform financial transactions like transfers to other people (P2P), payment transactions, and depositing and withdrawing funds (Suri, 2017a).

Mobile money, a type of e-money that does not require individuals to have an active bank account, had become a disruptive innovation capable of providing financial services at a lower cost than traditional banking, particularly in developing countries (Suri, 2017a). In 2019, there were 290 mobile money services in 95 countries with more than one billion registered accounts (GSMA, 2019). 50% of these services are in Sub-Saharan Africa. Additionally, Latin America and the Caribbean reported 27 mobile money services with 26 million registered accounts. Hence, the pandemic presented an opportunity to scale up e-money initiatives to deliver economic aid and promote financial inclusion.

2.4 Relevance of financial inclusion

Financial inclusion is a long-due goal in international development. According to the World Bank (2022), financial inclusion is understood as the capacity of individuals and firms to access and use affordable financial products and services in a responsible and sustainable way. Financial

inclusion is not a policy goal itself but an enabler for 7 of the 17 United Nations Sustainable Development Goals.

In this sense, access to financial services allows people to have a safe and convenient way to save their money, send and receive remittances from family and friends, and pay for goods and services (Demirgüç-Kunt et al., 2022). Serving the unbanked through digital solutions might have essential effects by pushing innovative, inclusive, and sustainable growth and reducing inequality. Addressing this market failure have collateral effects on the base of the pyramid, particularly in their participation in long-term investment activities, an efficient allocation of resources, lowering the cost of capital, smoothing consumption after unexpected shocks, and enhancing household finances (Beck & Demirgüç-Kunt, 2008; Demirgüç-Kunt et al., 2015, 2020).

Since the positive effects of financial inclusion are not directly derived from the ownership of accounts but from their appropriate and consistent use (World Economic Forum, 2018), it is vital to analyze if the pandemic and G2P initiatives have been translated into higher usage and economic empowerment. Therefore, for the effects of this paper, we will focus on measures of usage.

In the case of Colombia, financial inclusion is still a work in progress. Although the country has made considerable progress in increasing the share of the adult population with a savings account in the last decade, this share is still well below the average in developing economies and the world. By 2021, in Colombia, 60% of the adult population (older than 15 years old) reported having a savings account, compared to only 30% in 2011 (Demirgüç-Kunt et al., 2022). However, the average share of the adult population with savings accounts in developing economies is 71% and globally is 76% in 2021.

2.5 IS implementation.

Ingreso Solidario (IS), a social assistance payment program in Colombia, became a relevant case study of government and private firms coming together to provide financial support and enhance financial inclusion. The program was created in April 2020 with the primary goal of delivering financial aid to the most vulnerable households, without any previous support from the government. Furthermore, it delivered monthly 160.000 COP as monetary transfers (USD¹36.36; or USD 123.35 PPP²) without any conditions for recipients and targeted more than three million households (Gallego et al., 2021). According to a document by the Fiscal Rule Autonomous Committee of Colombia (CARF by its acronym in Spanish)³, the total national government spending in the program for the years 2020, 2021, and 2022 is 22.6 billion COP (USD 17.4 billion PPP). The program relied on 21 traditional banks and four digital wallets to transfer the money in a safe and fast way.

The program's official release was set in two waves (DNP, 2022). The first wave focused on households that already had a bank account (banked households). These were 1.16 million households that received their first disbursement in their savings accounts between the 7th and 10th of April 2020. The second wave aimed to reach 1.8 million households with no savings accounts in financial institutions (unbanked households). This group of individuals was assigned

¹ August 31st, 2022, exchange rate 1USD=4,400COP

² We use the World Bank Purchasing Power Parities factor of 2020: 1,297.1. See <https://data.worldbank.org/indicator/PA.NUS.PPP?locations=CO>

³ The document can be found in the following link: https://www.carf.gov.co/webcenter/ShowProperty?nodeId=%2FConexionContent%2FWCC_CLUSTER-201795%2F%2FidcPrimaryFile&revision=latestreleased

to one of four digital wallets⁴ depending on the individual's cellphone technology and the coverage of the company's local agents' presence in the municipality.

IS eligibility is mainly determined by the cut-off score based on the Sistema de Identificación de Potenciales Beneficiarios de Programas Sociales - SISBEN (Identification System for Potential Social Program Beneficiaries). However, there are four more conditions that individuals need to fulfill to receive IS financial aid. These conditions are detailed in section 5.1 *RD design* and are the main reason that explains why our design is a fuzzy RD with users that are identified as “compliers”, “never takers”, and “always takers”.

SISBEN is a multi-dimensional proxy means test of validated information for almost 40 million people, implemented through a national survey of income and assets. It determines if one qualifies for many of Colombia's social protection programs (Rosero et al., 2012). The estimation methodology of SISBEN has been updated and adjusted at different times since its inception in 1995 to improve its accuracy. The latest version is SISBEN IV.

SISBEN IV classifies individuals into four major groups:

- Group A: Individuals considered to be in extreme poverty. This group is sectioned into five levels, starting from A1 to A5, where individuals in group A1 have the lowest capacity to generate income.
- Group B: Individuals in poverty. The group is segmented into seven levels from B1 to B7, where higher levels mean better economic conditions.

⁴ The four digital wallets used to deliver the monetary transfers are *Daviplata*, *Nequi*, *Ahorro a la mano* and *Movii*

- Group C: Individuals in economic vulnerability. The group is sectioned into 18 levels from C1 to C18, where higher levels mean better economic conditions.
- Group D: No poor and no vulnerable people. The group is segmented into 21 levels from D1 to D21, where higher levels mean better economic conditions.

IS program targeted economic aid to individuals in SISBEN IV's levels ranging from A1 to C5 (DPS, 2022). After signing an NDA, the Department of National Planning (DNP) shared a continuous SISBEN score proxy variable. This proxy variable was normalized to have the cut-off value centered at zero.

Most unbanked households received IS financial aid in the form of e-money deposited in one of the four digital wallets authorized by the Colombian Government. This study focuses on IS unconditional e-money transfers through the digital wallet MOVii, the only branchless Fintech in the program. This company shared the data of its users' transactions and end-of-the-month balances.

3. Literature review

E-money has been used by policymakers in developing countries as a tool to achieve financial inclusion goals even before the pandemic. Suri (2017a) argues that e-money increases financial inclusion by improving access and enabling individuals to store their money using their phones. E-money also decreases dramatically the cost and time required for every transaction, making it a convenient alternative for individuals.

However, only access to an account is not enough. Sakyi-Nyarko et al. (2022) show that promoting the usage of e-money has a strong effect on well-being. The theory of change that explains such an effect is based on several mechanisms: E-money has shown a significant role in boosting welfare gains through improvements in risk-sharing, food security, consumption

smoothing, business profitability savings, access to essential services like healthcare and nutrition, and a better channel to disburse G2P programs than cash (Ahmed & Cowan, 2021; Aker et al., 2016; Aker & Mbiti, 2010; Danquah & Iddrisu, 2018; Jack & Suri, 2014; Riley, 2018; Suri, 2017b; Suri et al., 2021; Suri & Jack, 2016). Besides the aforementioned effects, using e-money instead of cash has the potential to substitute credit scoring and reduce asymmetric information, empowering women financially and assisting SMEs in growing in the formal sector (Davidovic et al., 2020). This means that e-money is the gateway to other financial services, such as credit and insurance, with even more direct effects on welfare.

In the case of Colombia, recent studies have studied the IS program and the role of digital wallets. Gallego et al. (2021) evaluated the impact of IS program on financial inclusion. The authors found that the program increased by 14 p.p. the probability of opening a bank account in treated individuals, compared to the control group, for individuals close to the cut-off value. Also, they found that IS-eligible households have a 7.5 p.p. higher chance of using digital wallets to pay their bills and do other transactions.

On the other side, Londoño-Vélez and Querubín (2022) found opposite effects for a similar unconditional cash transfer program to refund the VAT tax paid by low-income households. The intervention provided a sum lump transfer of USD 19 (USD 56.6 at PPP in 2019) delivered as hard cash or e-money. Furthermore, the authors found that surveyed individuals who received economic support as e-money reported no benefits. They would instead cash out their money as soon as possible to purchase goods and services.

This paper provides new evidence of e-money usage effects on financial inclusion in Colombia. First, contrary to recent studies in the same jurisdiction using self-reported information from surveys, this study is the first to use high-frequency administrative information on real-time

transactions done by individuals who received economic aid from IS. Additionally, the data enables us to isolate the effects of e-money on digital transactions and savings in a causal framework.

4. Data description

We use two main datasets that contain information on MOVii users' financial transactions and SISBEN IV information. The first dataset comes from the digital wallet MOVii company. This data is structured as a monthly panel dataset, where we observe if the user did wire money transfers or/and cash-out transactions in any month. Also, we have the user's savings account balance at the end of the month and the amount and date when they received financial aid from a government program. The unbalanced panel contains a total of 53 million observations. For each user, we observe up to 38 months of financial transactions, depending on the user's account creation date. The panel ranges from October 2019 to November 2022.

The second data set comes from the Department of National Planning (DNP). This data contains the SISBEN IV categorical score (e.g., A1, B4, ..., C3), a continuous proxy of the SISBEN IV score that ranges from 0 to 1, that we use as our running variable. The data also includes sociodemographic information such as date of birth, sex, and an identifier for receiving financial aid from other conditional cash transfer programs such as *Familias en Accion*, *Colombia Mayor*, *Jovenes en Acción*, or *Devolución del IVA (VAT refund)*.

After merging these two datasets, our data contains 9.2 million observations from 370,090 users. However, our study focuses on users that created their accounts between October 2019 and May 2020 (the reasons are explained in the next section). Therefore, our final sample contains 115,025 individuals with a total of 3.5 million observations. 44.4% receive IS monetary transfer, 46.5% receive no monetary aid and 9.1% receive financial aid from other sources than IS.

5. Methodology: RD design and estimation strategy

5.1 RD design

In order to estimate the causal impact of IS monetary transfer on financial inclusion, we use the SISBEN IV cut-off to implement an RD design. Since SISBEN IV score is a categorical variable, DNP shared with us a continuous variable that is a proxy to the SISBEN IV score. However, for practicality, we are going to call this proxy the SISBEN IV score.

Users with a SISBEN IV score above the cut-off are eligible to receive IS monetary transfer ($E_i = 1(SISBEN_i > c)$), where c is the cut-off for the assignment of IS financial aid. Then, the distance from user's SISBEN IV score to the cut-off is calculated, so the value of the cut-off is centered to zero ($c = 0$). Nevertheless, being SISBEN eligible is not enough to receive IS monetary transfer.

There are more criteria that individuals must fulfill to receive IS financial aid (DPS, 2022). First, none of the household members can receive financial aid from other government programs such as *Familias en Acción*, *Colombia Mayor* and *Jóvenes en Acción*. Second, they cannot be beneficiaries of the health-social security system. Third, they need to be identified as in monetary poverty. And fourth, they cannot hold savings products with more than 5 million COP (USD 3,854 PPP). However, only the first criteria are observed in our data.

Therefore, our RD design has compliers and non-compliers on both sides of the cut-off. We can identify the users that effectively received IS financial aid in our dataset by exploiting MOVii records on users' financial aid deposits. We denote $IS_i = 1$ if an individual received IS financial aid and $IS_i = 0$ otherwise. The eligible individuals that received the treatment "treated compliers" $\Pr(IS_i = 1|E_i = 1)$ represent 37.6% of our sample. Eligible individuals that did not receive the treatment "never takers" $\Pr(IS_i = 0|E_i = 1) = 34.5\%$. Non-eligible individuals that

received the treatment “always takers” $\Pr(IS_i = 1|E_i = 0) = 6.8\%$. Finally, the non-eligible individuals that did not receive the treatment “not-treated compliers” represent 21.1% - $\Pr(IS_i = 0|E_i = 0)$.

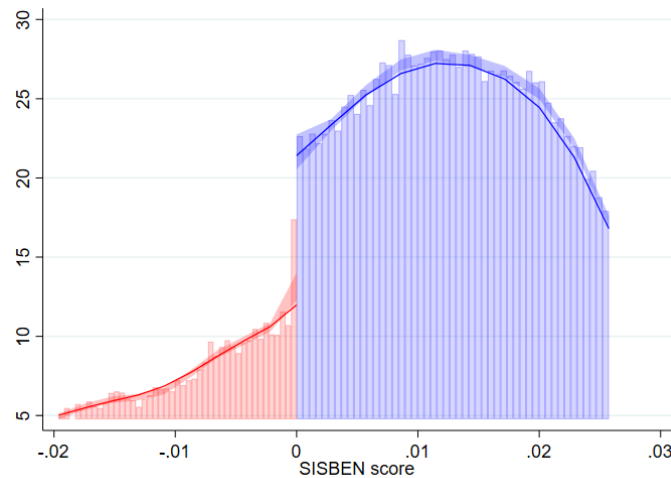
Following Cattaneo et al. (2023), we use a RD estimation strategy to estimate the effect of assigning IS on financial inclusion, or the Intention-to-Treat (ITT) effect. Also, we estimate the effect of receiving the treatment with a fuzzy RD, which can be interpreted as the average treatment effect for the subpopulation close to the cut-off point or Local Average Treatment Effect (LATE).

In order for an RD design to be valid, it must satisfy the following four assumptions (Cattaneo et al., 2019): (a) No evidence of manipulation of the assignment to the treatment; (b) observed differences around the cut-off are due only to differences in the running variable (the exclusion restriction); (c) the discontinuity at the cut-off causes a substantial change in the treatment assignment probability as a function of the assignment variable. (d) the relationship between the running variable and the outcome is consistent, either increasing or decreasing as the running variable crosses the cut-off point (monotonicity). We address each of these assumptions in the following paragraphs.

In the case of the first assumption, our study considers only the cohort of users that created their accounts at MOVii between October 2019 and May 2020 to exploit the randomness in the assignment of the treatment. Before the first disbursement of IS program in April 2020, very few people knew about the program. After the start of the program, individuals that wanted to apply to the program needed to ask for a SISBEN IV evaluation or re-evaluation, in case they already had a score that did not make them eligible to receive the financial aid and demonstrate that they fulfilled all the requirements. This process could take a few months. Therefore, individuals that

started to receive IS monetary transfers in the first two months of the program (April and May 2020) were less likely to change their scores by behaving strategically to receive the financial aid.

Figure 1. RD Density Test (2018)



Notes: The RD density test proposed by Cattaneo et al. (2018) rejects the null hypothesis of no manipulation with a robust corrected p-value of 0.0000.

However, the formal test to prove no manipulation of the assigning variable by Cattaneo et al. (2018) rejects our hypothesis of no manipulation. This local polynomial density estimator compares the densities on both sides of the cut-off and evaluates that there is no discontinuity in the polynomial function. In our case, there is a considerable jump in the distribution at the cut-off point, as can be observed in Figure 1. One hypothesis to explain this jump is due to the possibility that some users and their acquaintances created their account for the first time to receive the IS monetary transfer and use the digital wallet app. We must highlight that the first cohort of treated individuals between April and May 2020 received an SMS on their cell phones requiring them to download a determined app to open an account to access the monetary transfer. This announcement created a massive account opening phenomenon with about 87% of the accounts in our sample created between April and May 2020. Therefore, these accounts were created by IS treated

individuals and not-treated individuals that are, in many cases, eligible individuals ($E_i = 1(SISBEN_i > c)$), but they were not treated because they did not comply with all the requirements.

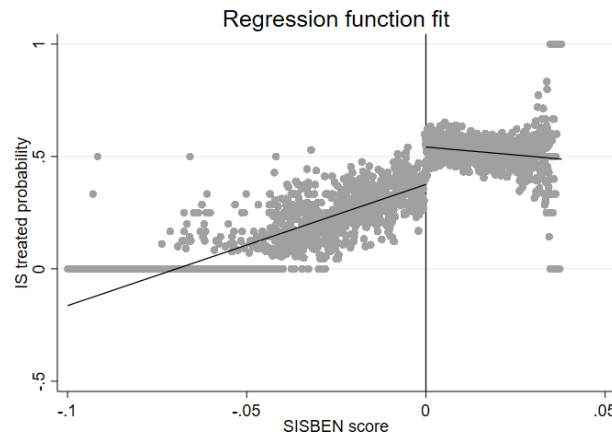
Despite not having the same densities on both sides of the cut-off, we are confident that the users had little chance to have had influenced their treatment assignment because the cohort of users included in our study were SISBEN evaluated months before the start of the program. Hence, we are confident that the estimation strategy is valid.

About the second assumption, in order for the exclusion restriction to be satisfied, it's expected that outcomes of interest are correlated with their eligibility status (E_i) only through its correlation with receiving the treatment. Therefore, we test individuals' observable covariates on either side of the cut-off to see if there are significant differences. In the case of female sex participation difference on both sides of the discontinuity, we find a coefficient of 0.0068 with a p-value of 0.485. Hence, no significant difference in user sex. And for the case of age, the difference is of 2.5 years more for users above the cut-off, which is significant at the 1% level. Although significant, age difference is relatively small in magnitude. In Table A1 in the appendix we report the RD regression results with these two covariates as outcomes and the SISBEN score as the running variable. In Figure A4 and A5 in the appendix, we complement these results with the graphical evidence that these covariates do not jump discretely at the cut-off in the case of the sex covariate and have a small jump in the case of the age covariate. Additionally, in Table A5, we present the share of women and average age for each group of users in our sample (eligible-complier, never-taker, always-taker, not-eligible-compliers).

We validate the third assumption by regressing the IS financial aid take-up against the running variable, SISBEN IV score, that is $E[IS_i | SISBEN_i]$. Figure 2 presents the discontinuity in the take-up rate at the cut-off point and the actual regression estimates a jump of 13 percentage

points (p.p.), increasing the probability of receiving the aid from about 40% to 53%. The discontinuity is significant at the 1% level, and it has a narrow 95% robust confidence interval between 0.0974 and 0.1539. The estimation uses a local linear regression at each side of the cut-off and a triangular kernel function.

Figure 2. Probability discontinuity at the cut-off point



Notes: The Figure presents the discontinuity on the probability of receiving the IS financial aid. The probability jumps from about 40% to 53%, an increase of 13 p.p. The estimation has a p-value of 0.000. The order polynomial fit is one (linear regression) and the kernel function is triangular.

In the case of the monotonicity assumption in our design, this is a plausible assumption. Since the treatment provides a monthly income in the form of e-money, the treatment consistently encourages the use of financial services such as wire money transfers and deposit accounts. On the other hand, users will steadily reduce their use of cash-outs as more people use digital wallet apps and become more confident in the use of these new technologies, as argued in the next section about technology adoption. Although we cannot test this assumption directly, the fact that our study analyses five consecutive semesters allow us to check consistency in the effects for each period.

Additionally, we present the results of a placebo test for each of our outcomes of interest, specifying artificial cut-offs (Cattaneo et al., 2019). To prevent the possibility of "contamination" caused by actual treatment effects, we exclusively utilize treated observations for the artificial cut-

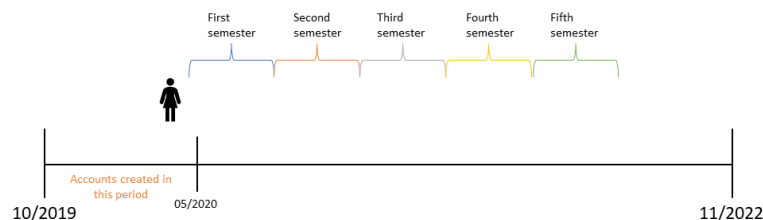
off that surpass the real cut-off. Conversely, we rely solely on control observations for the artificial cut-off that falls below the real cut-off. In this way, we guarantee that the placebo cut-off analysis uses observations with the same treatment status. Therefore, there should be no treatment effect by construction. The results can be validated in Tables A6-A8.

5.2 The technology adoption curve.

According to the technology-adopting life cycle, it takes time for high-tech markets to reach the main street. Thus, users can take a few months to actively use a digital wallet due to lack of trust or complexity of new technologies. Therefore, our identification strategy exploits the data's temporality to control for each user's digital wallet adoption curve period.

We select the users that created their accounts between October 2019 and May 2020 and run an RD design for the following five consecutive semesters they used the app. This strategy allows us to exploit the wide temporality of our data and test the adoption curve of this new technology. We implement this strategy to evaluate the effects on user's wire money transfers, cash-out, and end-of-the-month balances.

Figure 3. Five semesters evaluated



Notes: The Figure illustrates the five semesters analyzed in our study. This is a dynamic type of analysis that starts the moment the user opens her account at MOVii. We collapsed the information for each semester following the opening of the account. We include only users that created their accounts between October 2019 and May 2020.

6. RD results

For the estimation of the following results, we use the `rdrobust` package proposed by Calonico et al. (2014). In this nonparametric estimation, we specify linear regressions at both sides of the cutoff in order to not overfit our data. Also, we employ a triangular kernel function because we want to assign more weight to observations closer to the cut-off point.

We believe the main mechanisms in action that explain the following results are the recently acquired knowledge about these new technologies. Before the implementation of this program, people had low exposure to digital financial services. In some sense, the fact that they were obliged to start using these platforms to receive monetary aid and satisfy the distancing measures was an opportunity for them. This mechanism explains the results for each of the three outcomes of interest that we present: Wire money transfers, cash-out, and savings account balances.

6.1 Wire money transfers.

In our data, we observe if the user made use of the wire money transfer service in the app for each month, since the user created an account with MOVii. Using the RD estimation strategy to estimate the Intention-to-Treat (ITT) effect, we find an average effect in the five semesters tested of 1.4 p.p. This effect, compared to the average mean of the control group, 5.8%, means an increase of 24% in the probability of using the wire money transfer service. In Table 1 columns (1) to (5), we present the estimates for each semester, and in Figure 4 (left figure), we plot the tendency of the effect in the period analyzed. The highest effect was found in the third semester, 2.6 p.p., and the lowest effect was in the fifth semester, 0.83 p.p. For all semesters, we find a significant discontinuity. In the first four semesters, the discontinuity is significant at the 1% level, and in the fifth semester the discontinuity is significant at the 5% level.

Furthermore, the Fuzzy RD estimation strategy estimates the effect of receiving the treatment in the subpopulation near the cut-off – LATE. We find an average effect of 12.6 p.p., this implies an increase of 217% in the probability of using the wire money transfer service for the subpopulation near the discontinuity compared to the control group's average usage of wire money transfers. In Table 1 columns (6) to (10), we present the estimates for each semester, and in Figure 4 (right figure), we plot the tendency.

Figure 4. ITT and LATE semester coefficients for transfers

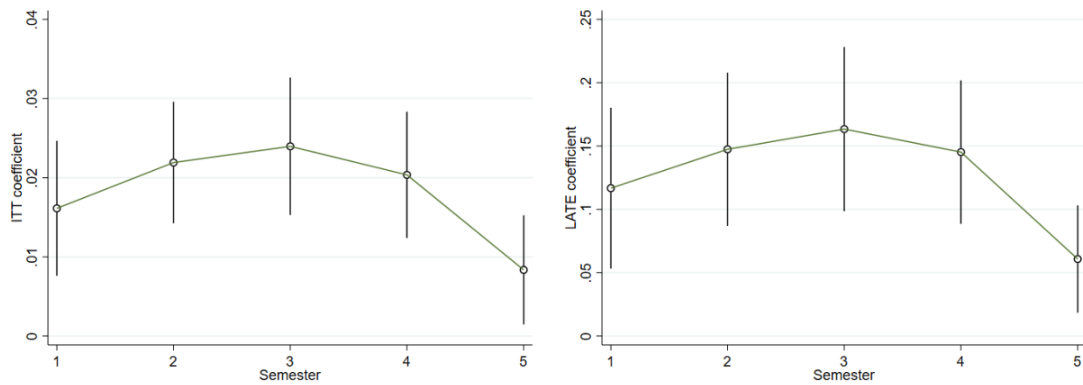


Table 1: ITT and LATE results for transfers

	ITT					LATE				
	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Coeff.	.0161	.0219	.0240	.0204	.0083	.117	.147	.163	.145	.0608
	(.004)	(.003)	(.004)	(.004)	(.003)	(.032)	(.030)	(.032)	(.028)	(.021)
P-value	.000	.000	.000	.000	.017	.000	.000	.000	.000	.005
M-C	.06	.06	.067	.059	.044	.06	.06	.067	.059	.044
N	113,576	113,222	112,603	112,215	111,984	113,576	113,222	112,603	112,215	111,984
BW	.011	.014	.013	.012	.009	.009	.010	.011	.010	.011

Notes: The table presents the effects of IS on the probability of using the wire money transfer service in the MOVii app per semester. From columns (1) to (5), the coefficients correspond to the ITT effect. Columns (6) to (10) present the effect for the subpopulation near the cut-off - LATE. Column (1) shows an effect of 1.6 percentage points increase in the probability of using the service. The basis is 6%, which means an increase of 26.7% growth in the probability of using the wire money transfer service. The RD estimation is run with the rdrobust package (Calonico et al., 2014). Robust standard errors in parentheses. M – C stands for mean of the control group. N is the number of observations. BW stands for bandwidth. The coefficients of the first stage of the Fuzzy RD regressions can be found in Table A2 in the appendix.

6.2 Cash-out.

Cash-out transaction translates into lower financial inclusion since individuals withdraw their money to use cash instead. As we expected, we observe a trend where treated users during the first semesters cash out their money rather often than the control users. However, the trend is progressively reduced while users start adopting this new technology and take advantage of the services offered by the digital wallet. The ITT and the LATE estimations follow a similar trend.

In the case of the ITT, the average effect for the five semesters is a 4.7 p.p. increase in the probability of cashing money out. However, the key finding here is the reduction in the trend to cash out. In the first semester, treated users have a 6.4 p.p. increase, while in the fifth semester, the increase is only 1.5 p.p. This means a reduction of 76.6% in the probability increase to cash out their money. Therefore, there is strong evidence of technology adoption from individuals above the cut-off point. The specific effects per semester can be found in Table 2 in columns (1) to (5). All the coefficients are significant at the 1% level. The trend is presented in Figure 5 (left figure).

On the other hand, the results for the LATE effect follow the same pattern as in the ITT estimation but with higher magnitudes. In particular, the average five semesters effect for the treated subpopulation in the discontinuity neighborhood is a 33.9 p.p. increase in the probability of cashing money out. Nevertheless, the semester trend is a reduction in this probability. By the fifth semester, the increase in the probability is only 11 p.p. The right figure in Figure 5 plots the medium-term effect of the program on the probability of cashing money out of the digital wallet. The effects for each semester are reported in Table 2 in columns (6) to (10).

Figure 5. ITT and LATE semester coefficients for cash-out

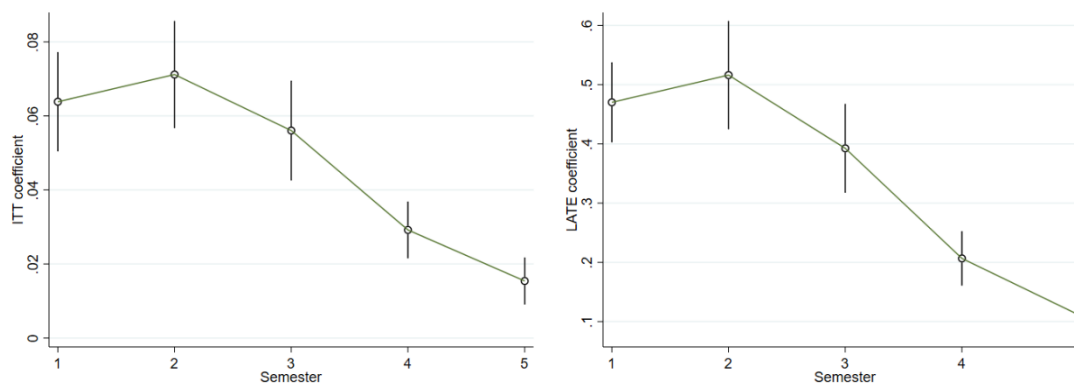


Table 2. ITT and LATE results for cash-out

	ITT					LATE				
	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Coeff.	.0638	.0712	.0560	.0292	.0154	.470	.516	.393	.207	.111
	(.006)	(.007)	(.006)	(.003)	(.003)	(.034)	(.046)	(.037)	(.023)	(.020)
P-value	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
M-C	.126	.136	.106	.065	.046	.126	.136	.106	.065	.046
N	113,576	113,222	112,603	112,215	111,984	113,576	113,222	112,603	112,215	111,984
BW	.011	.011	.010	.012	.011	.010	.007	.009	.011	.010

Notes: The table presents the effects of IS on the probability of cashing money out of the MOVii app per semester. From columns (1) to (5), the coefficients correspond to the Intention-to-Treat (ITT) effect. Columns (6) to (10) present the effect for the subpopulation near the cut-off – LATE. Column (1) shows an effect of 6.4 p.p. increase in the probability of cashing out. The control mean is 12.6%, which means an increase of 50.8% growth in the probability of cashing money out of the app. The RD estimation is run with the *rdrobust* package (Calonico et al., 2014). Robust standard errors in parentheses. M – C stands for mean of the control group. N is the number of observations. BW stands for bandwidth. The coefficients of the first stage of the Fuzzy RD regressions can be found in Table A3 in the appendix.

6.3 Savings Account Balance

Our data allow us to observe the user's balance at the end of each month. This figure presents an idea of how much money is left in the user account after covering all her obligations. Even though this variable is not a perfect measure of user saving behavior, it's a good approximation to understand if the user is using the digital wallet account to store her money and her ability to react to income shocks.

In the case of the ITT effect, we find that the five-semester average effect is an increase of \$5,735 COP (4.42 USD PPP) in the end-of-the-month balance. This number represents a 27.6% increase with respect to the control five-semester average end-of-the-month balance, \$20,812 COP (16 USD PPP). The lowest increase was found in the second semester, \$2,991 COP (2.3 USD PPP), which implies an increase of 12.6% compared to the control average end-of-the-month balance during the second semester. The fourth semester reported the highest rise, with a jump of \$7,711 (5.9 USD PPP), an increase of 44.8% with respect to the control group's balance that semester.

The LATE estimation shows the following estimates for the subpopulation near the cut-off. The five-semester average effect is \$40,504 COP (USD 31 PPP) increase in the end-of-the-month balance. This represents 25% of IS monetary transfer. The effects trend for the five semesters evaluated is plotted in the right figure of Figure 6. The coefficients are reported in Table 3 in columns (6) to (10). The coefficients of the first stage can be found in Table A4 in the Appendix.

Figure 6. ITT and LATE for the balance

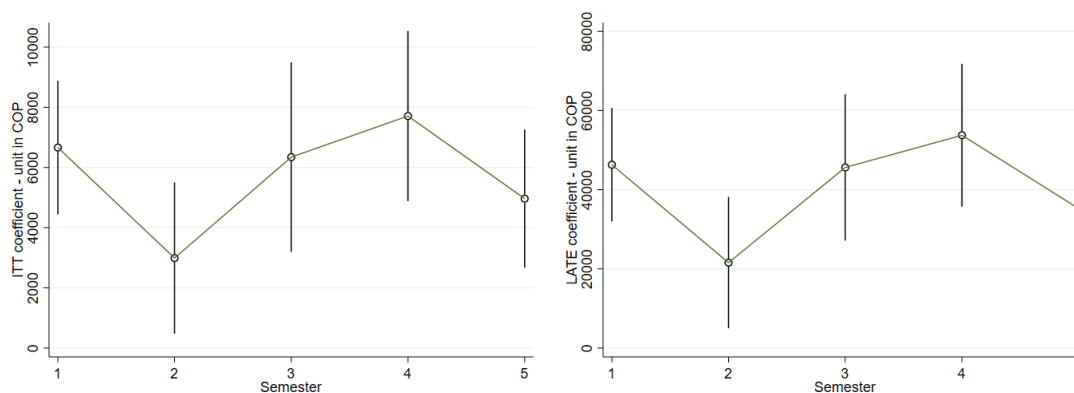


Table 3. Sharp and Fuzzy RD results for the end-of-the-month balance

	ITT					LATE				
	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Coeff.	6,664	2,991	6,345	7,711	4,966	46,315	21,576	45,626	53,761	35,246
	(1,124)	(1,273)	(1,598)	(1,435)	(1,164)	(7,263)	(8,396)	(9,367)	(9,146)	(8,093)
P-value	.000	.019	.000	.000	.000	.000	.010	.000	.000	.000
M-C	19,685	23,746	28,209	17,214	15,210	19,685	23,746	28,209	17,214	15,210

N	112,442	112,128	111,484	111,094	110,866	112,442	112,128	111,484	111,094	110,866
BW	.010	.009	.008	.008	.013	.009	.009	.009	.010	.012

Notes: The table presents the effects of IS on the average end-of-the-month balance in the MOVii account per semester. From columns (1) to (5), the coefficients correspond to the Intention-to-Treat (ITT) effect. Columns (6) to (10) present the effect for the subpopulation near the cut-off – LATE. Column (1) shows an effect of \$6,664 COP increase in the average end-of-the-month balance. The control mean balance is \$19,685, which means an increase of 33.9% growth in the average end-of-the-month balance. The RD estimation is run with the rdrobust package (Calonico et al., 2014). Robust standard errors in parentheses. M – C stands for mean of the control group. N is the number of observations. BW stands for bandwidth. The coefficients of the first stage of the Fuzzy RD regressions can be found in Table A4 in the appendix.

7. Conclusions

This work reflects the importance and effectiveness of new technologies to tackle long-lasting social issues. An exogenous shock, such as the COVID-19 pandemic, allowed public and private parties to come together with a solution that went beyond of providing resources for vulnerable people to mitigate poverty and became a successful mechanism to promote financial inclusion.

Promoting financial inclusion can be a platform to overcome structural challenges embedded in our societies, such as no or low access to capital, education, and health care. That's why it is exciting and motivating to find innovative solutions that not only bring access to financial products but also encourage their usage. However, these solutions take time to witness structural changes as their adoption in daily life activities is a slow process. Our work provides evidence of these changes, changes that can present almost immediate results, as in the case of wire money transfers and end-of-the-month balance, or can take time to start showing positive results, as in the case of cash-out transactions.

We expect this research paper to contribute to the discussion of the importance of effective financial inclusion in the path to reaching the United Nations' 17 Development Goals. Given the effect of IS on financial inclusion, a plausible policy recommendation is to further expand this type of program.

The fact that these users employed basic financial tools might not have a great impact on their wellbeing. Nonetheless, transactional tools are a gateway to more complex financial products such as loans, insurance, investments, etc with direct effects on social welfare. However, this is just the start. Sometimes the hardest part is this part. Now that people, who never before had any financial products, are starting to use these basic financial tools, they will have greater chances of using more complex products that will enable them to have access to capital for their next business idea, education for their kids, or even primary health care.

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Appendix

Table A1. Sex and Age covariates

	Sex (1)	Age (2)
Coeff	.0067 (.009)	2.46 (.39)
P-value	.485	.000
Z	0.69	6.23
N	113,576	113,576
BW	.014	.008

Table A2. Wire Money Transfers. LATE – First stage

	LATE – Wire Money Transfers				
	S1 (1)	S2 (2)	S3 (3)	S4 (4)	S5 (5)
Coeff – FS	.1388 (.011)	.1418 (.011)	.1429 (.011)	.1425 (0.11)	.1415 (.010)
P-value	.000	.000	.000	.000	.000
Z	11.68	12.57	12.93	12.68	13.06
N	113,576	113,222	112,603	112,215	111,984
BW	.009	.010	.011	.010	.011

Table A3. Cash-out. LATE – First stage

LATE – Cash-out				
S1	S2	S3	S4	S5

	(1)	(2)	(3)	(4)	(5)
Coeff – FS	.1411 (.011)	.1266 (.013)	.1414 (.011)	.1419 (.010)	.1422 (.011)
P-value	.000	.000	.000	.000	.000
Z	12.25	9.60	11.93	13.26	12.33
N	113,576	113,222	112,603	112,215	111,984
BW	.010	.007	.009	.011	.010

Table A4. End-of-the-month Balance. LATE – First stage

LATE – End-of-the-month balance					
	S1	S2	S3	S4	S5
	(1)	(2)	(3)	(4)	(5)
Coeff – FS	.1405 (.011)	.1414 (.011)	.1420 (.011)	.1403 (.011)	.1404 (.010)
P-value	.000	.000	.000	.000	.000
Z	11.78	11.88	11.86	12.32	13.39
N	112,442	112,128	111,484	111,094	110,866
BW	.009	.009	.009	.010	.012

Table A5. Sex and Age average

	Sex (1=female)	Age
	(1)	(2)
All sample	.65	40
Treated-compliers $\Pr(IS_i = 1 E_i = 1)$	.74	43
Never-takers $\Pr(IS_i = 0 E_i = 1)$	.66	38
Always-takers $\Pr(IS_i = 1 E_i = 0)$	.64	43
Not-treated-compliers $\Pr(IS_i = 0 E_i = 0)$	.50	36

Table A6: Placebo test results for transfers

	Artificial cut-off = 0.03					Artificial cut-off = -0.03				
	S1 (1)	S2 (2)	S3 (3)	S4 (4)	S5 (5)	S1 (6)	S2 (7)	S3 (8)	S4 (9)	S5 (10)
Coeff.	.0096 (.008)	.0221 (.008)	.0176 (.009)	.0164 (.007)	-.0023 (.006)	-.0041 (.003)	-.0076 (.003)	-.0049 (.004)	-.0046 (.004)	-.0089 (.002)
Robust P-value	.397	.015	.188	.137	.497	.463	.053	.487	.629	.015
N	51,052	50,712	50,178	50,006	49,959	62,524	62,510	62,425	62,209	62,025
BW	.010	.014	.010	.009	.008	.008	.008	.007	.006	.010

Notes: The table presents the placebo test results on the probability of using the wire money transfer service in the MOVii app per semester. From columns (1) to (5), the coefficients correspond to the placebo test specifying the artificial cut-off = 0.03. For this test, we use only treated observations. Columns (6) to (10) present the placebo results for the artificial cut-off = -0.03. For this test, we rely on not-treated individuals. The RD estimation is run with the rdrobust package (Calonico et al., 2014). Robust standard errors in parentheses. N is the number of observations. BW stands for bandwidth.

Figure A1. Placebo test results – artificial cut-off = 0.03 (left figure) and artificial cut-off = -0.03 (right figure) – Transfers

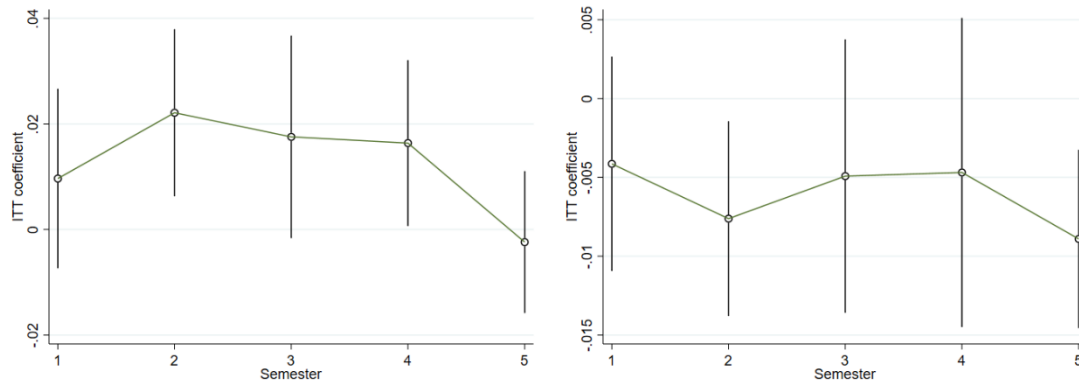


Table A7: Placebo test results for cash-outs

	Artificial cut-off = 0.03					Artificial cut-off = -0.03				
	S1 (1)	S2 (2)	S3 (3)	S4 (4)	S5 (5)	S1 (6)	S2 (7)	S3 (8)	S4 (9)	S5 (10)
Coeff.	.0108 (.012)	.0096 (.012)	.0106 (.010)	.0060 (.005)	-.0059 (.004)	-.0036 (.001)	-.0053 (.001)	-.0059 (.005)	-.0049 (.004)	-.0101 (.002)
Robust P-value	.245	.308	.548	.539	.325	.029	.020	.460	.452	.000
N	51,052	50,712	50,178	50,006	49,959	62,524	62,510	62,425	62,209	62,025
BW	.007	.008	.009	.011	.014	.006	.006	.007	.008	.011

Notes: The table presents the placebo test results on the probability of cashing money out of the MOVii app per semester. From columns (1) to (5), the coefficients correspond to the placebo test specifying the artificial cut-off = 0.03. For this test, we use only treated observations. Columns (6) to (10) present the placebo results for the artificial

cut-off = -0.03. For this test, we rely on not-treated individuals. The RD estimation is run with the rdrobust package (Calonico et al., 2014). Robust standard errors in parentheses. N is the number of observations. BW stands for bandwidth.

Figure A2. Placebo test results – artificial cut-off = 0.03 (left figure) and artificial cut-off = -0.03 (right figure) – Cash-out

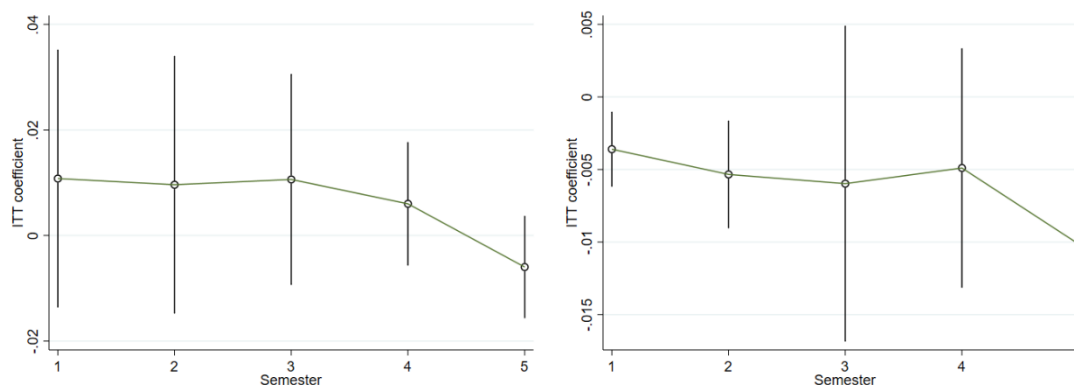


Table A8. Placebo test results for the balance

	Artificial cut-off = 0.03					Artificial cut-off = -0.03				
	S1 (1)	S2 (2)	S3 (3)	S4 (4)	S5 (5)	S1 (6)	S2 (7)	S3 (8)	S4 (9)	S5 (10)
Coeff.	2,482 (2,271)	-4,236 (2,563)	-4,841 (2,528)	5,021 (3,297)	-3,922 (3,505)	-870.9 (575.0)	-396.5 (701.7)	-107.8 (959.9)	1,032 (897.1)	499.9 (613.6)
Robust P-value	.475	.067	.154	.300	.193	.296	.828	.841	.468	.552
N	50,546	50,223	49,679	49,520	49,460	61,945	61,898	61,801	61,602	61,405
BW	.009	.010	.010	.008	.008	.006	.007	.009	.010	.013

Notes: The table presents the placebo test results on the average end-of-the-month balance in the MOVii app per semester. From columns (1) to (5), the coefficients correspond to the placebo test specifying the artificial cut-off = 0.03. For this test, we use only treated observations. Columns (6) to (10) present the placebo results for the artificial cut-off = -0.03. For this test, we rely on not-treated individuals. The RD estimation is run with the rdrobust package (Calonico et al., 2014). Robust standard errors in parentheses. N is the number of observations. BW stands for bandwidth.

Figure A3. ITT and LATE for the balance

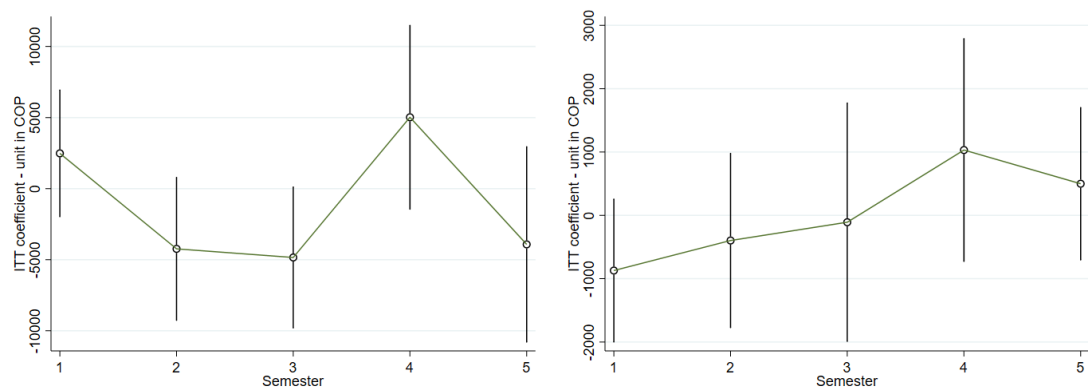


Figure A4. Graphical illustration of the regression discontinuity between covariate sex and
SISBEN score



Figure A5. Graphical illustration of the regression discontinuity between covariate age and
SISBEN score

